THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 35	$x^3+y^3=(x+y)^3-3xy(x+y)$.
2	**(a)** 98	$x^3-y^3=(x-y)^3+3xy(x-y)$.
3	**(c)** 18	$x^3+\frac{1}{x^3}=\left(x+\frac{1}{x}\right)^3-3\left(x+\frac{1}{x}\right)$.
4	**(b)** 1030301	$101=100+1$; use $(a\pm b)^3$.
5	**(c)** 21	$2xy=(x+y)^2-(x^2+y^2)$.
6	**(b)** 28	$2xy=(x^2+y^2)-(x-y)^2$.
7	**(d)** 16	Add 2 to both sides.
8	**(d)** $(2x+3)(4x^2-6x+9)$	Use the factorisation identity.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$103^2=(100+3)^2=10000+600+9=10609$, using R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A follows from $a^2-b^2=(a-b)(a+b)$; R is true but is a different identity and does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^3+y^3=(x+y)^3-3xy(x+y)$ (1); $=512-288=224$ (1).	1+1
12	$103=100+3$ (1); $103^3=1092727$ (1).	1+1
13	Use $a^3+b^3=(a+b)(a^2-ab+b^2)$; $a=2x$, $b=1$ (1); $=(2x+1)(4x^2-2x+1)$ (1).	1+1
14	$x^3+y^3=(x+y)^3-3xy(x+y)$ (1); $=1000-480=520$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=7^3-3(7)$ (1); $=322$ (1).	1+1+1
16	Need two numbers with sum -5 and product 4; They are -1 and -4 (1); $=(x-1)(x-4)$ (1); At $x=6$: $(5)(2)=10$ (1).	1+1+1
17	Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=2x$, $b=3$ (1); $=(2x-3)(4x^2+6x+9)$ (1).	1+1+1
	OR: $\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=8^3-3(8)$ (1); $=488$ (1).	1+1+1
18	Need two numbers with sum -9 and product 14; They are -2 and -7 (1); $=(x-2)(x-7)$ (1); At $x=10$: $(8)(3)=24$ (1).	1+1+1
	OR: Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=3x$, $b=1$ (1); $=(3x-1)(9x^2+3x+1)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(100+3)^2=10000+600+9=10609$ (1). (ii) $(100-2)^2=10000-400+4=9604$ (1). (iii) $(100+2)(100-2)=10000-4=9996$ (1). (iv) $(1000-3)(1000+3)=1000^2-3^2=999991$ (1) using $(a-b)(a+b)=a^2-b^2$ (1).	1+1+1+2
20	$x^2+y^2=4+48=52$ (1); $x^3-y^3=(x-y)^3+3xy(x-y)=8+144$ (1); $x^3-y^3=152$ (1); $(x+y)^2=(x-y)^2+4xy=4+96$ (1); $(x+y)^2=100$ (1).	1+1+1+1+1
	OR: $2xy=(x+y)^2-(x^2+y^2)=81-45=36$ (1); $xy=18$ (1); $(x-y)^2=x^2+y^2-2xy=45-36=9$ (1); $x-y=3$ (1); $x^2-y^2=(x+y)(x-y)=27$ (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $(a+b)^2=a^2+b^2$ and leaving out the middle term $2ab$.
- Using $(a-b)^2=a^2-b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3-3a^2b+3ab^2-b^3$.
- Not squaring the whole term: $(2x+3)^2=4x^2+12x+9$, not $2x^2+12x+9$.
- Factorising x^2+5x+6 with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.

- Not taking out a common factor before applying an identity, e.g. $2x^2 - 18 = 2(x - 3)(x + 3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 52	$x^3+\frac{1}{x^3}=\left(x+\frac{1}{x}\right)^3-3\left(x+\frac{1}{x}\right)$.
2	**(c)** 970299	$99=100-1$; use $(a\pm b)^3$.
3	**(b)** 14	$2xy=(x+y)^2-(x^2+y^2)$.
4	**(b)** 24	$2xy=(x^2+y^2)-(x-y)^2$.
5	**(b)** 25	Add 2 to both sides.
6	**(a)** $(3x-2)(9x^2+6x+4)$	Use the factorisation identity.
7	**(b)** 72	$x^3+y^3=(x+y)^3-3xy(x+y)$.
8	**(c)** 37	$x^3-y^3=(x-y)^3+3xy(x-y)$.
9	**(c)** A is true but R is false.	A is the correct identity; R is false because the middle term $-2xy$ is missing.
10	**(d)** A is false but R is true.	A is false (take $a=b=1$: $8\neq 2$); R is the correct identity.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$97=100-3$ (1); $97^3=912673$ (1).	1+1
12	Use $a^3+b^3=(a+b)(a^2-ab+b^2)$; $a=3x$, $b=2$ (1); $=(3x+2)(9x^2-6x+4)$ (1).	1+1
13	$x^3+y^3=(x+y)^3-3xy(x+y)$ (1); $=216-90=126$ (1).	1+1
14	$105=100+5$ (1); $105^3=1157625$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=9^3-3(9)$ (1); $=702$ (1).	1+1+1
16	Need two numbers with sum -8 and product 15; They are -3 and -5 (1); $=(x-3)(x-5)$ (1); At $x=9$: $(6)(4)=24$ (1).	1+1+1
17	Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=x$, $b=6$ (1); $=(x-6)(x^2+6x+36)$ (1).	1+1+1
	OR: $\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)$ (1); $x^3+\frac{1}{x^3}=10^3-3(10)$ (1); $=970$ (1).	1+1+1
18	Need two numbers with sum -11 and product 18; They are -2 and -9 (1); $=(x-2)(x-9)$ (1); At $x=12$: $(10)(3)=30$ (1).	1+1+1
	OR: Use $a^3-b^3=(a-b)(a^2+ab+b^2)$ (1); $a=5x$, $b=2$ (1); $=(5x-2)(25x^2+10x+4)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x+4$ m (1). (ii) $(x+4)^2=x^2+8x+16$ m^2 (1). (iii) $(x+4)^2-x^2=8x+16$ m^2 (1). (iv) $8(12)+16=112$ m^2 (1); cost $=112\times 50=$ Rs 5600 (1).	1+1+1+2
20	$x^2+y^2=16+90=106$ (1); $x^3-y^3=(x-y)^3+3xy(x-y)=64+540$ (1); $x^3-y^3=604$ (1); $(x+y)^2=(x-y)^2+4xy=16+180$ (1); $(x+y)^2=196$ (1).	1+1+1+1+1
	OR: $2xy=(x+y)^2-(x^2+y^2)=169-89=80$ (1); $xy=40$ (1); $(x-y)^2=x^2+y^2-2xy=89-80=9$ (1); $x-y=3$ (1); $x^2-y^2=(x+y)(x-y)=39$ (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $(a+b)^2=a^2+b^2$ and leaving out the middle term $2ab$.
- Using $(a-b)^2=a^2-b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3-3a^2b+3ab^2-b^3$.
- Not squaring the whole term: $(2x+3)^2=4x^2+12x+9$, not $2x^2+12x+9$.
- Factorising x^2+5x+6 with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2-18=2(x-3)(x+3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 27	$2xy = (x+y)^2 - (x^2+y^2)$.
2	**(b)** 12	$2xy = (x^2+y^2) - (x-y)^2$.
3	**(c)** 36	Add 2 to both sides.
4	**(b)** $(x-4)(x^2+4x+16)$	Use the factorisation identity.
5	**(c)** 189	$x^3+y^3 = (x+y)^3 - 3xy(x+y)$.
6	**(c)** 136	$x^3-y^3 = (x-y)^3 + 3xy(x-y)$.
7	**(a)** 198	$x^3 + \frac{1}{x^3} = \left(x+\frac{1}{x}\right)^3 - 3\left(x+\frac{1}{x}\right)$.
8	**(a)** 941192	$98 = 100 - 2$; use $(a \pm b)^3$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$49 - 20 = 29$, obtained by using R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A follows from $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ with $a = 2x,\ b = 3$; R is true but is not used to prove A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use $a^3+b^3 = (a+b)(a^2-ab+b^2)$; $a = x,\ b = 5$ (1); $= (x+5)(x^2-5x+25)$ (1).	1+1
12	$x^3+y^3 = (x+y)^3 - 3xy(x+y)$ (1); $= 1331 - 792 = 539$ (1).	1+1
13	$95 = 100 - 5$ (1); $95^3 = 857375$ (1).	1+1
14	Use $a^3+b^3 = (a+b)(a^2-ab+b^2)$; $a = 4x,\ b = 3$ (1); $= (4x+3)(16x^2-12x+9)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\left(x+\frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x+\frac{1}{x}\right)$ (1); $x^3 + \frac{1}{x^3} = 11^3 - 3(11)$ (1); $= 1298$ (1).	1+1+1
16	Need two numbers with sum -7 and product 6; They are -6 and -1 (1); $= (x-6)(x-1)$ (1); At $x = 8$: $(2)(7) = 14$ (1).	1+1+1
17	Use $a^3-b^3 = (a-b)(a^2+ab+b^2)$ (1); $a = 4x,\ b = 1$ (1); $= (4x-1)(16x^2+4x+1)$ (1).	1+1+1
	OR: $\left(x+\frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x+\frac{1}{x}\right)$ (1); $x^3 + \frac{1}{x^3} = 12^3 - 3(12)$ (1); $= 1692$ (1).	1+1+1
18	Need two numbers with sum -8 and product 16; They are -4 and -4 (1); $= (x-4)(x-4)$ (1); At $x = 9$: $(5)(5) = 25$ (1).	1+1+1
	OR: Use $a^3-b^3 = (a-b)(a^2+ab+b^2)$ (1); $a = 2x,\ b = 5$ (1); $= (2x-5)(4x^2+10x+25)$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(2x+1)^3 = 8x^3+12x^2+6x+1$ (1). (ii) Edge $= 5$ cm, so volume $= 125$ cm^3 (check: $64+48+12+1 = 125$) (1). (iii) $1000+300+30+1 = 1331$ (1). (iv) $(3x-2)^3 = 27x^3-54x^2+36x-8$ (1); at $x = 1$: $27-54+36-8 = 1 = 1^3$ (1).	1+1+1+2
20	$x^2+y^2 = 1+84 = 85$ (1); $x^3-y^3 = (x-y)^3 + 3xy(x-y) = 1+126$ (1); $x^3-y^3 = 127$ (1); $(x+y)^2 = (x-y)^2 + 4xy = 1+168$ (1); $(x+y)^2 = 169$ (1).	1+1+1+1+1
	OR: $2xy = (x+y)^2 - (x^2+y^2) = 121 - 61 = 60$ (1); $xy = 30$ (1); $(x-y)^2 = x^2+y^2-2xy = 61-60 = 1$ (1); $x-y = 1$ (1); $x^2-y^2 = (x+y)(x-y) = 11$ (1).	1+1+1+1+1

Mistakes to avoid:

- Writing $(a+b)^2 = a^2+b^2$ and leaving out the middle term $2ab$.
- Using $(a-b)^2 = a^2-b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3-3a^2b+3ab^2-b^3$.
- Not squaring the whole term: $(2x+3)^2 = 4x^2+12x+9$, not $2x^2+12x+9$.

- Factorising $x^2 + 5x + 6$ with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2 - 18 = 2(x - 3)(x + 3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND